

ECON 108 Discussion Section

Week 2. Simple Linear Regression Model: Practice Questions

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Question 1.

$$\text{Test score}_i = \beta_0 + \beta_1 \text{STR}_i + u_i$$

Suppose we are interested in examining the effect of class size, measured by the student-teacher ratio (str), on average test scores (testscr). The following information is based on school district-level data.

. su testscr str

Variable	Obs	Mean	Std. dev.	Min	Max
Y testscr	420	654.1565 $\bar{Y}$	19.05335 S_Y	605.55	706.75
X str	420	19.64043 $\bar{X}$	1.891812 S_X	14	25.8

. corr testscr str, cov
(obs=420)

	testscr	str
Y testscr	363.03 $\rightarrow S_Y^2$	
X str	-8.15932 $\text{cov}(X, Y)$	3.57895 $\rightarrow S_X^2$

	X_1	X_2
X_1	$\text{cov}(X_1, X_1) = V(X_1)$	
X_2	$\text{cov}(X_1, X_2)$	$\text{cov}(X_2, X_2) = V(X_2)$

Let's calculate the estimated coefficients manually.

- a. What is $\hat{\beta}_1$? $\hat{\beta}_0$? Interpret $\hat{\beta}_1$.

$$\hat{\beta}_1 = \frac{\text{cov}(X, Y)}{S_X^2} = \frac{-8.159}{3.79} = -2.28$$

When a school district's STR increases by one student per class, test scores decrease by 2.28 points.

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 654.157 - (-2.28)(19.64) = 698.936$$

- b. What is R^2 ? Interpret the meaning of the number.

$$R^2 = (r)^2 = (-0.226)^2 = 0.051$$

$$r = \frac{\text{cov}(X, Y)}{S_X \cdot S_Y} = \frac{-8.159}{(1.892)(19.053)} = -0.226$$

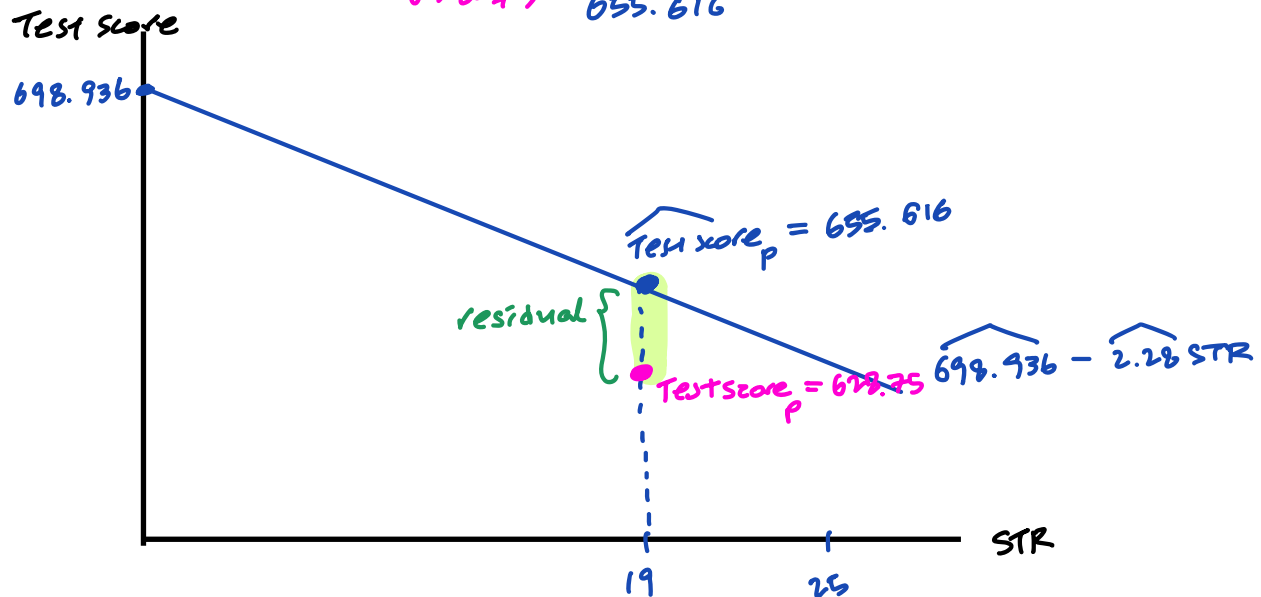
$$\text{STR} = 19$$

c. Predict testscr if the student-teacher ratio (str) is 19.

$$\begin{aligned} \widehat{\text{Test score}} &= \hat{\beta}_0 + \hat{\beta}_1 \text{STR} \\ &= 698.936 - 2.28 \text{STR} \\ &= 698.936 - 2.28 (19) \\ &= 655.616 \end{aligned}$$

d. The student-teacher ratio for Perris School District is 19, and the average test score is $\checkmark 628.75$. In the graph below, draw the estimated regression line and label the following for Perris School District:

- STR
- Actual average test score
- Predicted average test score
- residual = $628.75 - 655.616$



Did we calculate everything correctly?

```
. reg testscr str
```

Source	SS	df	MS	Number of obs	=	420
Model	7794.11004	1	7794.11004	F(1, 418)	=	22.58
Residual	144315.484	418	345.252353	Prob > F	=	0.0000
				R-squared	=	0.0512
				Adj R-squared	=	0.0490
Total	152109.594	419	363.030056	Root MSE	=	18.581

testscr	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
str	-2.279808 $\hat{\beta}_1$	.4798256	-4.75	0.000	-3.22298	-1.336637
_cons	698.933 $\hat{\beta}_0$	9.467491	73.82	0.000	680.3231	717.5428

```
. margins, at(str=19)
```

Adjusted predictions

Number of obs = 420

Model VCE: OLS

Expression: Linear prediction, predict()

At: str = 19

	Delta-method					
	Margin	std. err.	t	P> t	[95% conf. interval]	
_cons	655.6166	.9573181	684.85	0.000	653.7348	657.4984

Question 2.

We are also interested in the relationship between the number of computers and average test scores. The regression results are presented below, with standard errors shown in parentheses.

$$\widehat{TestScore}_i = \frac{\hat{\beta}_0}{(1.127)} - \frac{\hat{\beta}_1}{(0.002)} Computer_i \quad (1) \quad \checkmark$$

$$\widehat{TestScore}_i = \frac{643.36}{(2.08)} - \frac{79.40}{(13.81)} Computer \text{ per Student}_i \quad (2)$$

- a. Interpret the estimated coefficients. (Choose one model.)

Slope $\hat{\beta}_1$: Additional computer in a school district decreases test scores by 0.003, all else equal.

Intercept $\hat{\beta}_0$: ^{when} a school district has zero computers, predicted test scores are 655.122.

H_0

H_a

- b. Conduct a significance test at the 5% level to evaluate whether there is any relationship between the number of computers and average test scores. Show all your steps. Be sure to show how you calculate the necessary probabilities from a z table.

① | $H_0: \beta_1 = 0 \equiv \beta_{1,0}$ ← no relationship btw #comp & test scores
 $H_a: \beta_1 \neq 0$ ← there is a relationship



② sig lv (α) = 0.05

③ test statistic = $\frac{\hat{\beta}_1 - \text{mean of } \hat{\beta}_1 \text{ under } H_0}{\text{sd of } \hat{\beta}_1 \text{ under } H_0}$

• mean of $\hat{\beta}_1$
 $= E(\hat{\beta}_1) = \beta_1$

under H_0 , $\beta_{1,0}$

$$= \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{se}(\hat{\beta}_1)}$$

• sd of $\hat{\beta}_1 \rightarrow \text{se}(\hat{\beta}_1)$

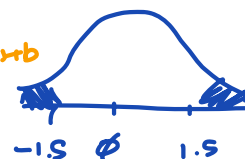
$$= \frac{\hat{\beta}_1 - 0}{\text{se}(\hat{\beta}_1)}$$

$$= \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)} = \frac{-0.003}{0.002} = \underline{\underline{-1.5}}$$

④ p-value $\Rightarrow P(|t| > |\text{test statistic}|)$

$$= P(|t| > 1.5)$$

t distb $\approx$ z distb
when sample
size is large!



$$= 2 \times P\left(\frac{z}{t} < -1.5\right)$$

0.0668 z-table!

$$= \underline{\underline{0.1336}}$$

⑤ Conclusion

p-value vs sig lv (α)

0.1336 > 0.05 $\Rightarrow$ fail to reject H_0

(no relationship
btw # computers
and test scores.)

- c. Construct a 95% confidence interval for the slope parameter. Discuss how the results are complementary to your findings in part b).

* not $\hat{\beta}_1$ 😞

A 95% CI for $\beta_1 = \hat{\beta}_1 \pm 1.96 \cdot se(\hat{\beta}_1)$

$$= -0.003 \pm 1.96 (0.002)$$

$$= -0.003 \pm 0.004$$

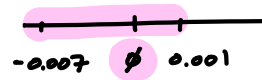
95% of the time

β_1 is w/in this interval

$$= [-0.003 - 0.004, -0.003 + 0.004]$$

$$= [-0.007, 0.001]$$

contains ϕ



$\therefore$ We cannot rule out that β_1 is zero.

no relationship btw # computers
and test scores

The actual regression results are shown below. How accurate were our manual calculations of the test statistic, p-value, and confidence interval (CI)?

. reg testscr computer

Source	SS	df	MS	Number of obs	=	420
Model	827.015554	1	827.015554	F(1, 418)	=	2.29
Residual	151282.578	418	361.920043	Prob > F	=	0.1314
				R-squared	=	0.0054
				Adj R-squared	=	0.0031
Total	152109.594	419	363.030056	Root MSE	=	19.024

testscr	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
computer	-.0031833	.0021058	-1.51	0.131	-.0073226	.0009561
_cons	655.1223	1.126888	581.36	0.000	652.9072	657.3374

test stat
Q6③

p-value
Q6④

Q3.

. reg testscr comp_stu

Source	SS	df	MS	Number of obs	=	420
Model	11146.6436	1	11146.6436	F(1, 418)	=	33.05
Residual	140962.95	418	337.231938	Prob > F	=	0.0000
				R-squared	=	0.0733
				Adj R-squared	=	0.0711
Total	152109.594	419	363.030056	Root MSE	=	18.364

testscr	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
comp_stu	79.40485	13.81145	5.75	0.000	52.2563	106.5534
_cons	643.3633	2.080229	309.28	0.000	639.2743	647.4523

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

